

Understanding the Roles of Geometric Forms and Proportions in CNN-Based Image Classification

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Abstract - The performance of Convolutional Neural Networks (CNNs) in image-based AI/ML applications can be affected by factors such as: number of training datasets used, geometric forms of subjects in images, and resizing for capturing geometric proportions. To the authors' best knowledge, adequate information is not available on the influence of some of these factors on the accuracy of image classification using a CNN driven by deep learning techniques. To address the stated lacunae, two custom datasets were designed: one that contains cars, trucks and roses to investigate the possible effect of geometric form variation, and a second case study consisting of the images of lionesses and kittens, both belonging to the cat family, to assess the impact of proportional resizing. To bestow confidence on the results obtained, the effects of training dataset size, and values of hyper-parameters on accuracy of image classification are also investigated. The results indicate that a natural form may be more difficult to classify as compared to man-made geometry, and maintaining geometric proportions can add to rate of success of image classification.

Keywords - Convolutional Neural Networks (CNNs), Image Classification, Dataset Size, Feature Diversity, Image Scaling, Preprocessing Techniques, Deep Learning

I. INTRODUCTION

Convolutional Neural Networks (CNNs) have altered the manner that can be used to classify images through learning and extracting features from the images [1], [2]. These networks are now indispensable components of such diverse industries as diagnostics of diseases using medical images, or object recognition and navigation in self-driving cars, to name a few. It also finds its application in facial recognition systems, to boost the security and authentication measures [3]. At its core, image analysis involves three interconnected tasks:

- *Image Identification*, as the very phrase implies, deals with answering if a given image such as a face of a person matches with any image present in a large pool of similar stored images of multiple subjects.
- *Image Classification* involves categorizing images into certain classes or categories using information of images of heterogeneous objects, such as deciphering if a given image is that of a car or a truck or a flower.
- *Feature Recognition* is the process of determining if any feature in a given image belongs to a certain category, in terms of - for example - pattern or texture, which may be required for applications such as medical diagnostics.

The current work focuses on Image Classification, which is a key process of making sense of visual data for use in real-world applications. The performance of CNNs in classification tasks hinges on several key factors. Firstly, Dataset Size can play a significant role. The quantity of data for training, validation, and testing gives us an indication of how well the model will perform on inputs that it has never 'seen' before.

The problem is that larger datasets lead to better feature representation but such databases are sometimes difficult to generate because of practical limitations [4]. Secondly, Feature Diversity is another critical factor. This is because variability within the dataset makes it difficult to classify the data. For instance, some of the categories may also share some characteristics which are often not well defined making it difficult for a model to learn from such sets [5]. Lastly, Image Processing also plays an essential role. Basic operations such as resizing, cropping or enlarging the images can greatly affect the performance of the model. Pre-processing should be done correctly to maintain uniformity and can even improve feature extraction [6]. Although benchmarks such as CIFAR-10 dataset [7], which includes 60,000 labeled images, are widely used to evaluate CNNs, such a large volume of data may not be available in many practical applications and guidelines may be needed for deciding upon an optimum number of training cases for a given problem. As an effort in this direction, the present investigation focuses on analyzing the effects of geometric forms (such as natural versus man-made) and proportions (such as a large animal in comparison with a smaller one of the same family) on CNN accuracy using two different case studies.

To the authors' best knowledge, there is scanty information in the public domain on the dependence of success of image classification on the factors mentioned. Although relative comparisons are of main importance here, a study has never-the-less been conducted on the influence of training database size, and values of various hyper-parameters such as limit on epochs, number of hidden layers and learning rate on image classification accuracy using available library functions. The current exposition is organized as follows: Section II reviews background and related work, including a discussion on CIFAR-10 and scaling approaches such as SiCNN [8] and ELASTIC [9] frameworks; Section III details the current methodology, covering data collection, pre-processing, model design, and evaluation techniques; Section IV presents the results and analysis of the experiments conducted on two custom datasets; Section V concludes the study with key findings and implications; finally, Section VI outlines directions for future research to enhance CNN performance in constrained scenarios.

II. BACKGROUND

The CIFAR-10 dataset referred in this study as a benchmark for image classification contains a total of 60000 labeled images as mentioned above, and ten different classes with 6000 images per class. This large and diverse dataset allows models to have good stability, and CNN models can generally obtain testing accuracy of about 86.5%. While this level of accuracy shows the efficacy of large datasets, it is not adequately understood as to the minimum number of datasets required for obtaining satisfactory assessments or predictions.

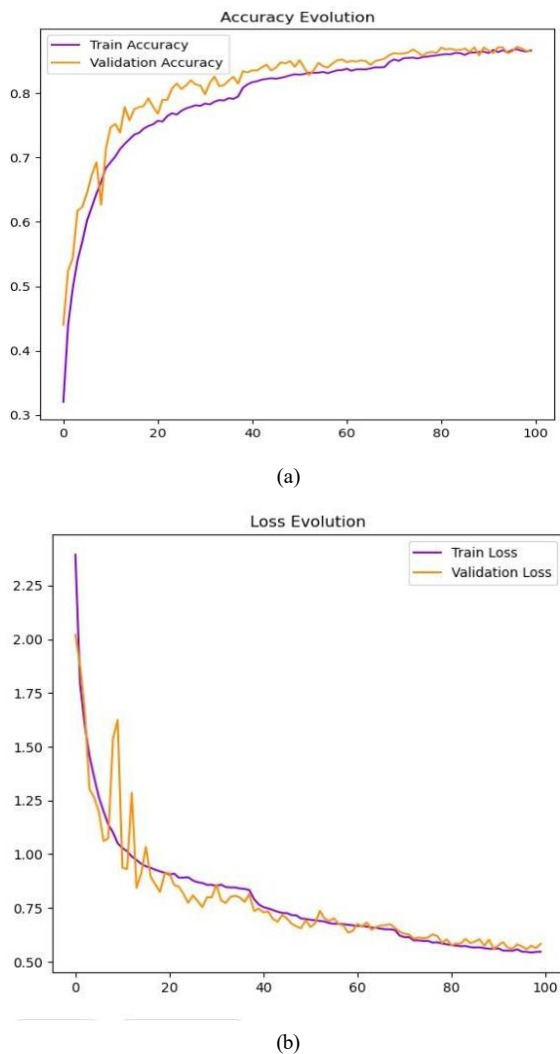


Fig. 1. (a) Accuracy vs. Epoch graph for CIFAR-10 dataset; (b) Loss vs. Epoch graph for CIFAR-10 dataset [7].

A graph illustrating the training and testing accuracy achieved on CIFAR-10 is shown in Fig.1. Apart from the number of data, scaling presents numerous challenges in CNN performance particularly when the objects in a dataset have different sizes. Two important investigations that attempt to address this problem are Scale-Invariant CNN (SiCNN) by Xu et al. [8] and ELASTIC framework by Wang et al [9]. To solve the issue of scale variance in image classification, Xu et al. introduced a multi-column architecture. In this method of using SiCNN, the input image is first divided into many sub images each of which is passed through a different convolutional column that is trained to look at features of a particular size. They are then combined in the last layers to make the predictions. Fig.2 shows an image of the SiCNN architecture proposed in the paper. This approach makes it easy for the model to learn features that are invariant to scale because the size of the object does not affect the model's ability to classify the object.

However, SiCNN brings certain difficulties in terms of computational complexity because the images are processed at multiple scales. Also, the architecture is designed for particular datasets, which makes it restricted to different tasks in terms of its application. SiCNN helps to increase accuracy in the cases of a large difference in the scale of the input data and filters. However, this approach does not solve all the

problems related to scaling in CNNs.

In contrast to earlier efforts, the ELASTIC framework is a dynamic system that does not rely on external up-sample and down-sample components, but incorporates them directly within specific CNN layers [9]. In contrast to static scaling where resizing is uniformly done on all images, ELASTIC can change scaling policies over time depending on the characteristics of the input data it has been fed on. This adaptive approach makes it possible to capture features at different scales in equal ability and achieve better accuracy in icon classes for datasets with different sizes of objects.

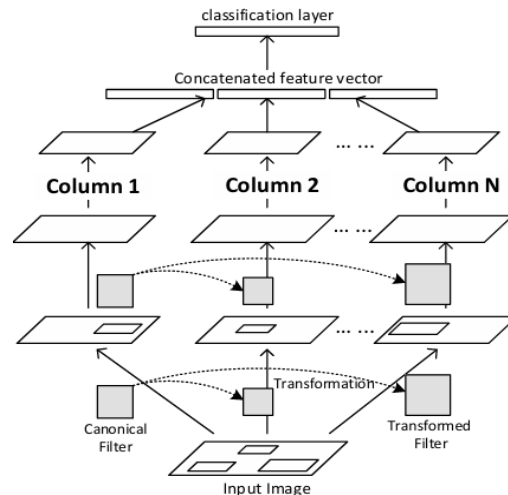


Fig. 2. SiCNN architecture [8]

In the work of Wang et al., it is shown that the proposed ELASTIC approach provides consistent gains over different datasets because of its ability to handle the issue of variability within a class caused by scale variations. However, such a framework's gist is based on dynamic scaling policies within the architectures that reduce its effectiveness. ELASTIC targets feature representation at multiple scales while not providing a solution to the question of scale invariance in CNN construction.

Nguyen et al. [10] analyzed the effect of the size of a training set in the classification of text data in supervised learning where the data collection contains short and noisy texts from Twitter. Their study systematically examined the impact of increasing data sizes on three Twitter-related tasks: including sentiment analysis, evaluative information detection measure, and information relevance measure. They described the difference between new deep learning models and old machine learning when measured by data inputs. The analysis showed that utilization of pre-trained models helps to cope with the small datasets problem. It is also crucial to select the appropriate text representation strategy for training the model and the model performance does not necessarily improve with the increasing of the size of the dataset. This work provides important recommendations for choosing learning models and allocating resources in supervised tasks.

Papoutsis et al. [11] compared recently proposed deep learning models for large-scale Land Use and Land Cover (LULC) image classification based on the BigEarthNet Sentinel-2 multispectral dataset. They experimented with 62 trained models which consist of CNN, Multi Layer Perceptrons (MLPs) and Vision Transformer

Perceptrons (MLPs) and Vision Transformer while evaluating few factors like classification accuracy, time taken to train a model and inference rate. The proposed approach was a scaled lightweight Wide Residual Network (WRN) with the Efficient Channel Attention mechanism resulting in 4.5% higher F-Score across 19 LULC classes compared to ResNet50, but with fewer parameters. On the diversified remote sensing tasks, they demonstrated that the presented models have transfer learning capabilities [18], including SEN12MS where the top WRN achieved only 7% off from state-of-art performance (71.1% of F-Score) while being trained on limited data.

Liu et al. [12] discusses deploying small deep learning models for ore image classification and how depth, structure of model or size of dataset affects them. By experimenting with six CNN architectures based on AlexNet and VGGNet, optimized with Batch Normalization (BN) layers, the study reveals key findings: deeper models are better suited for increasing the classification rate within a lower number of training vectors provided, and BN layers help to increase performance and minimize overfitting in cases of a low size of samples. This explains why dataset size affects model performance differently for various networks; whereas VGG19_BN gives better results with relatively smaller datasets, AlexNet_BN yields better predictions for large datasets. Methods such as Grad-CAM and Heat maps strengthen the fact that the models offer high efficiency in identifying the relevant features. The study evaluates the effectiveness of small CNNs in the classification of mineral images and future research on problems such as poor image quality as a result of the conditions in industries.

Dablain et al [13] addresses the challenges CNNs face when working with imbalanced image datasets, particularly their difficulty in generalizing to minority classes and the black-box nature of their decision-making process. The paper explores the latent features and class embedding (CE) of CNNs, hypothesizing that the ability of a model to generalize to minority classes is linked to the their top-K CE and feature embedding, which vary in magnitude depending on whether the data is balanced or imbalanced.

Through experiments with three popular image datasets and two cost-sensitive algorithms, the study finds that CNNs trained with imbalanced data struggle to generalize if the latent features' magnitudes do not align between the training and test sets. The results highlight the importance of class feature diversity and suggest that balancing the latent feature diversity, not just the class distribution, is crucial for improving performance. The study concludes that the diversity of latent features in minority classes is as important as the number of class examples for improving model performance. It emphasizes that current methods focusing on rebalancing class numbers and margins may not be sufficient and suggests future research into data augmentation techniques, visualization of latent features, and exploring new methods to disentangle class representations to improve generalization. The findings contribute to a better understanding of the behaviors of CNNs with imbalanced data and point towards areas for further investigation. Although all these methods usually enhance the accuracy in the presence of variable sized objects.

They do not provide complete size invariance and complicate the architectural design in several contexts. Pre-trained model architectures and good data representation as Nguyen et al. [10] pointed out that, massive datasets enhancement of the performance is not always guaranteed as the size of the corpus increases. Papoutsis et al. studied big image classification employing modest Wide Residual Networks by fewer parameters, while Liu et al. [12] established an idea that deeper models with batch norms outperforms, particularly in small datasets.

Dablain et al. [13] looked at how CNN performed on imbalanced datasets to focus on how class variations influencing minor class generalizations. However, all of these findings are targeted at specific datasets or use complicated structures, so there are issues with comprehensively considering scale and feature variations [17]. Therefore, to close these gaps, this study presents the datasets that capture real-world scaling, preprocessing methodologies of small and heterogeneous datasets, and a basic CNN model that performs well with small-scale datasets.

The above studies present the progress and the shortcomings encountered in addressing scaling issues in CNNs. Compared to other solutions, SiCNN and ELASTIC offer new approaches, while they both depend on certain architectures [16] or scaling mechanisms which have prompted the further search for a more general solution. The current work seeks to extend the reported studies by targeting, in particular, the effects of 'geometric form diversity (GFD)' and 'proportional resizing (PR)' on accuracy of classification by a CNN. It needs to be pointed out that GFD is not quite the same as 'feature diversity' mentioned earlier, and PR needs to be distinguished from the above scaling-related considerations as will be apparent later.

III. METHODOLOGY

Advancements in Convolutional Neural Networks (CNNs) have brought about vast improvements in image classification since they are capable of directly learning features. However, there is a strong dependency between their performance and the quality as well as characteristics of the data set files. The following section describes the experimental preparation where two new datasets, comprising (i) 'cars, trucks, and roses' and (ii) 'lionesses and kittens' were used. The main purpose was to investigate the effects of feature differentiation and scaling on an effective CNN classification. It may be noted that in dataset (i), cars and trucks are of man-made geometry with limited geometric profiles while roses being of natural forms can have infinite variations in shapes within the same overall class. In dataset (ii), it would be challenging for a CNN model to distinguish between lionesses and kittens in terms of geometric shapes alone as both types of animals belong to the cat family and possess strong resemblance in terms of features and geometric outlines. However, a main distinguishing factor is that lionesses have a consistent shade of yellowish grey body color while kittens can be of variegated coats. Additionally, an adult lioness is much larger in size as compared to an average kitten, and it this latter difference that is directionally captured in the current study through resizing of images prior to CNN-based analysis.

Code Structure and Functions

In order to use the CNN model for implementation as well

evaluation, several functions were utilized in the code. This basic framework remained consistent throughout the experiments with both datasets, ensuring a standardized structure.

The CNN structure applies four convolutional stages, each with increasing operational complexity. The first layer convolved the image with a 3×3 kernel, and the subsequent three layers perform similar operations with 3×3 kernels. Each convolutional layer is followed by a max-pooling layer to reduce spatial dimensions while preserving significant features. This hierarchical approach allows the model to discover fine details such as edges and global structures, which are critical for classifying heterogeneous datasets [15].

Various steps involved in the current methodology are shown in blocks in Fig. 3. Building on this structured workflow, the subsequent sections delve into each step of the methodology, linking it coherently to the overall study process.

Data Collection and Splitting

The datasets employed in this study were carefully selected to investigate feature variability and the performance of CNN based on the geometric and proportional resizing of images. Data augmentation was done using the ImageDataGenerator function, where transformations like rotation, zooming, and horizontal flips were applied. These augmentations added more variation to the training data, while standardizing the pixel values to be as close as possible to the mean. Dataset (i) consisted of three different classes where each one was represented by 300 images. These were divided into 250 images for training the model, 50 for model validation and a separate set of 10 images for testing. Dataset (ii) comprised of 300 images per class, with 240 images for training, 60 images for validation and the remaining 15 being test images. This was a balanced way of splitting into training and evaluation in addition to forming diversity in the type of data samples.

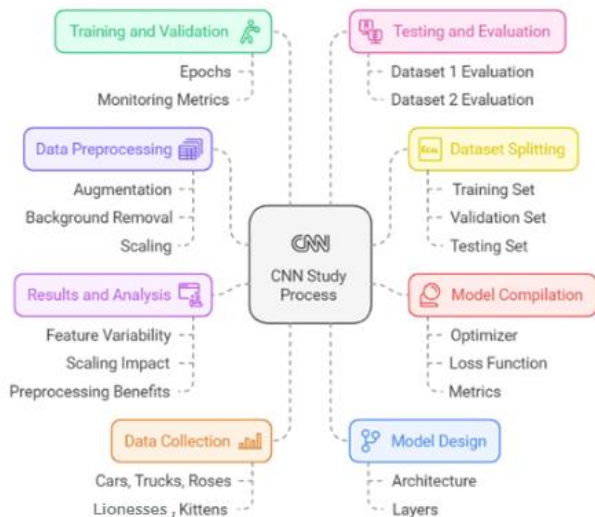


Fig. 3. A structured overview of the CNN study process.

Data Pre-Processing

Data pre-processing can be said to have particularly contributed most to the increase in the rate of accuracy of models. To augment a dataset, ImageDataGenerator was utilized which applied transformations such as rotation, zooming, and horizontal flips.

Apart from the steps mentioned previously, for dataset (ii) comprising lionesses and kittens, further pre-processing was

done to resize them to their natural proportions.

Unlike conventional scaling techniques that resize all images uniformly and are automated, the current approach is centred around adjusting the image sizes so that the relative geometric proportions of a lioness and a kitten is reflected in the training database. To the authors' best knowledge, no CNN or image data pre-processing algorithm is currently able to automatically achieve this attribute of capturing relative sizes of objects in images used for AI-based training.

The images of lionesses and kittens were reduced to a width and height measurement that reflected their proportional sizes as shown in Fig. 4. This resizing is done taking into account the real-life proportional lengths of both animals, with a lioness typically measuring around 5.15 feet in length and a kitten around 0.8 feet, resulting in an approximate longitudinal dimensional ratio of 6:1. It is noted that proportional resizing as explained here is not same as scaling as dwelt upon in published literature.

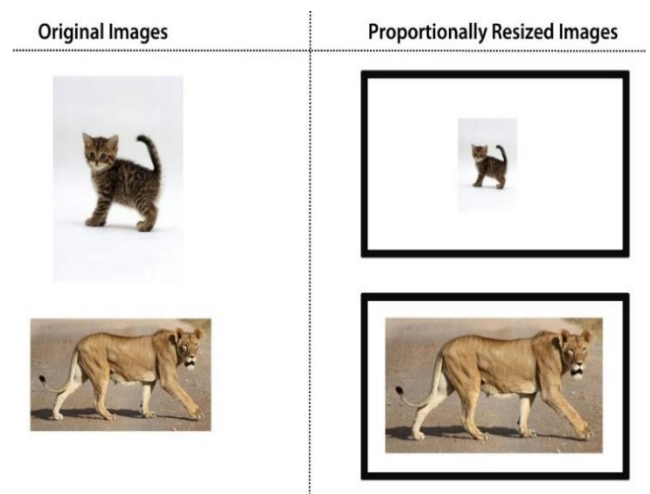


Fig. 4. Demonstration of 'proportional resizing': the original images (left) are resized to proportional sizes (right) based on real-life dimensions

Model Design and Compilation

The sequential CNN model was designed with four convolutional layers, each of increasing complexity, paired with max-pooling layers. The first convolutional layer used a 3×3 kernel, while the subsequent three layers also used 3×3 kernels. This hierarchical structure enabled the model to capture low-level features such as edges and textures while also learning high-level object structures. The final layers included a fully connected layer with ReLU activation and a softmax classifier to predict class probabilities.

Training and Validation

The model was compiled to work with the Adam [15] optimizer with a learning rate of 0.0005.

Testing and Evaluation

Accuracy and loss curves for the model were plotted with reference to epochs. These graphs were used during the training phase to determine whether the desired convergence was achieved and to ensure proper generalization of validated datasets. A utility function was developed to perform feature extraction and image classification on single images using the trained model. This function was then applied to test data, demonstrating that the model could handle real-world inputs without significant loss of accuracy.

IV. RESULTS AND ANALYSIS

For Dataset (i)

The first dataset aimed to examine the effect of variation of form/contour on CNN classification accuracy using three distinct classes viz. cars, trucks, and roses, each represented by 100 images. The unseen test sets of 10 images per class were formed to evaluate the prediction capability of the proposed model, while training and validation sets were used to set up the prediction network and verify its consistency. The training accuracy of 98.25% achieved with dataset (i) demonstrates a substantively superior performance comparable to the CIFAR-10 dataset, despite the significantly smaller size of training data, highlighting the robustness of the model. Fig. 5 represents the training accuracy (i.e. Accuracy vs. Epoch) and the Loss vs. Epoch graphs.

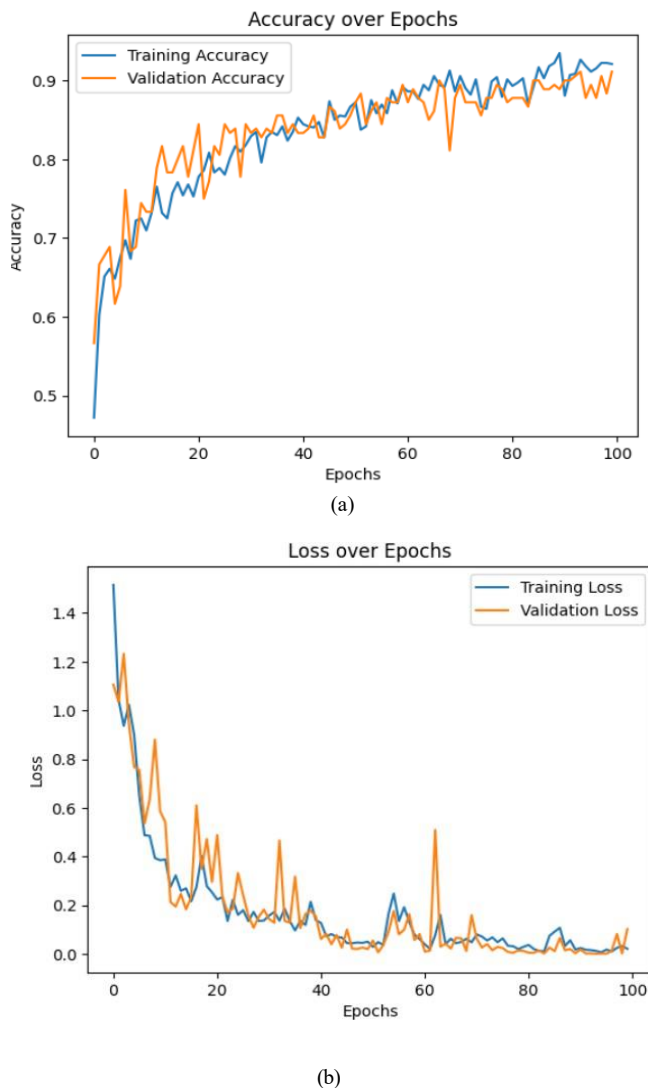


Fig. 5. (a) Accuracy vs. Epoch and (b) Loss vs. Epoch graphs for dataset (i) illustrating the model's convergence during training

The bar graph in Fig.6 illustrates the success rate of predictions across each of the three classes viz. trucks, cars, and roses.

The higher testing accuracy for trucks and cars vis-à-vis roses can perhaps be attributed to intrinsically less variation in geometric forms for the former two classes which are man-made and well-defined in comparison with roses which, due to their natural origin, have endless variations in contours and

configurations.

Fig. 7 provides further insights into the reasons for failure of current CNN to correctly classify images of roses in certain cases during testing. It is noticed in the latter figure that roses characterized by features such as being at the center of the picture, upright in orientation, and with shades of red matching typical RGB values, were correctly classified. However, roses that did not fit into these descriptions – for instance, those laid flat, off-centered, or having unconventional colors like green – were often misclassified. The failed cases in Fig. 7 tend to confirm the supposition that more variations can be expected in configurations of roses, although admittedly, further scrutiny is needed on this topic.

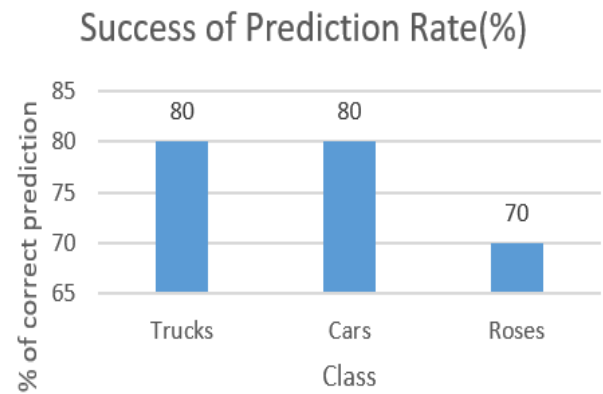


Fig. 6. Prediction accuracy (%) for dataset (i), showing higher accuracy for trucks and cars compared to roses due to form consistency

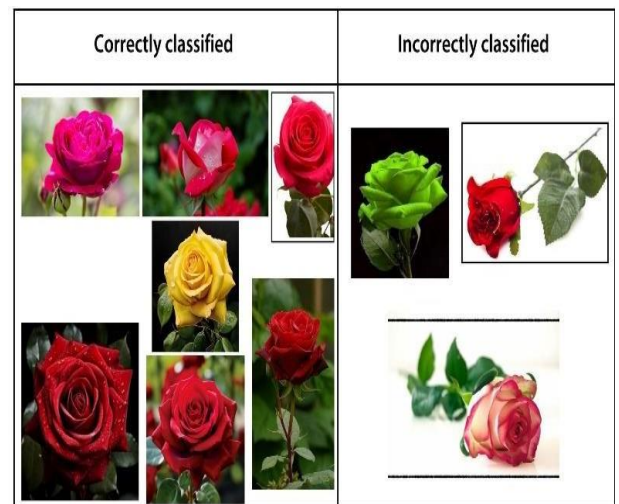


Fig. 7. Classified and misclassified images of roses

In view of the remark made above, an investigation is carried out on the influence of hyper-parameters on the results obtained here. In this context, the parameters that have been varied are: number of CNN layers (i.e. 2, 4 and 6), maximum limit on epochs (i.e. 80 or 100), and learning rate (i.e. 0.0010.005, 0.0001 and 0.0005). Additionally, batch size is also varied. The results of this study are summarized in Table 1. It is seen in the last row of Table 1 that values of training and testing accuracies are maximum for the values of parameters that have been chosen in the current study for which results have been reported above in Figs. 5 through 7.

No. of layers	Epochs	Learning Rate (LR)	Batch Size	Training Accuracy (%)	Testing Accuracy (%)	Testing Accuracy (%)	Testing Accuracy (%)
				(Overall)	(Roses)	(Trucks)	(Cars)
2 Conv Layers	100	0.0001	32	91.99	35	40	50
6 Conv Layers	100	0.0001	32	92.33	30	35	35
4 Conv Layers	80	0.005	16	88.88	45	65	60
4 Conv Layers	100	0.001	128	89.67	25	45	50
4 Conv Layers	100	0.0005	32	98.25	70	80	80

Table.1. Impact of various relevant parameters on classification accuracy

Additionally, it is noted that for all combinations of values of factors considered in Table 1, the testing accuracy for roses is consistently lower as compared to the corresponding testing accuracies for cars and trucks. As the number of images has been kept as constant for all classes of objects considered, it may be justifiable to say that the higher variation in form of roses has made it challenging for the current CNN to achieve the same degree of classification accuracy (i.e. 80% in Fig. 6) that has been reached for cars and trucks as compared to that (70% in Fig. 6) for roses.

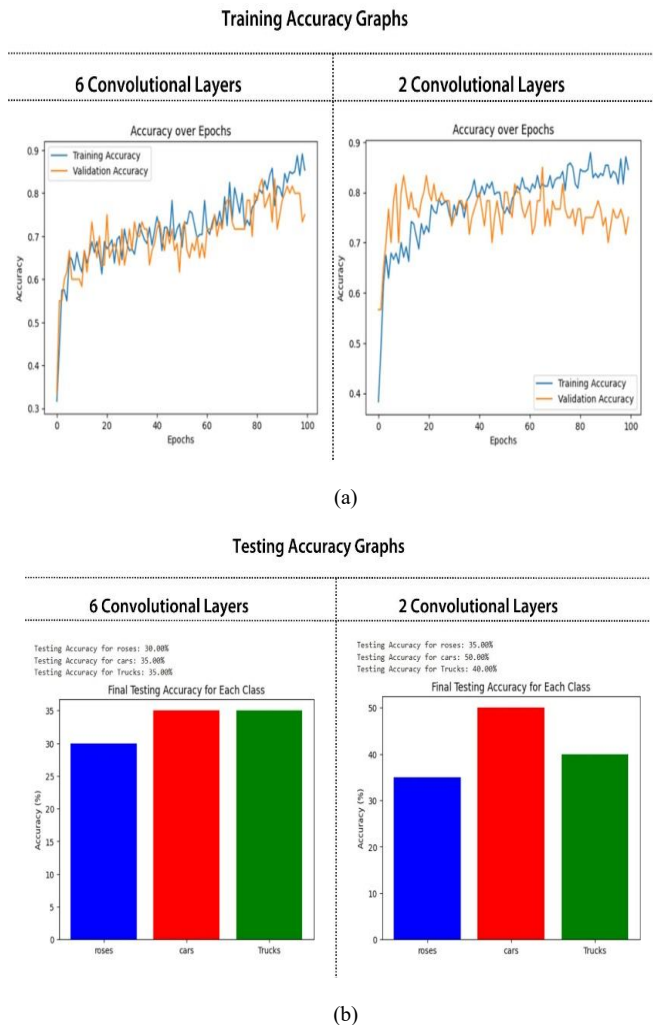


Fig. 8. (a) Training accuracy variations for different CNN layers; (b) corresponding testing accuracy bar charts

For further clarification, the results (related to training accuracy variation with respect to number of epochs, and testing accuracies) for the first two rows in Table 1 are shown graphically in Fig. 8. Furthermore, the impact of dataset size on training accuracy was evaluated and the results are

presented in the form of a graph in Fig. 9.

The consistently increasing trend of training accuracy with respect to training dataset size confirms that the current implementation of AI/ML for image classification using CNN is highly reliable. The maximum training accuracy of 98.2% has been reached for 300 images per class (i.e. a total of 900 images for three classes) for which inferences have been drawn here. It appears that a relatively miniscule dataset size of 900 (i.e. 300 per class) compared to a mammoth 60,000 images (i.e. 6000 per class with 10 classes) considered in CIFAR-10 indicates that number of images alone cannot yield high accuracy of image classification, but a reasonable dataset size needs to be supplemented with data that is of high quality in terms of representing forms meaningfully.

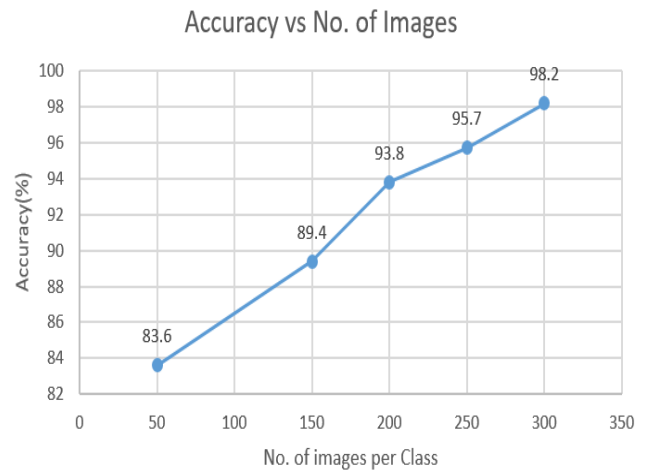
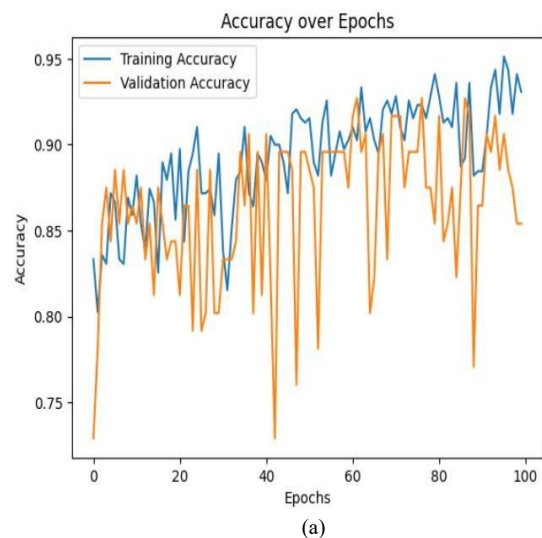


Fig. 9. Graph illustrating training accuracy vs. number of images per class in dataset (i) for training and validation

For Dataset (ii)

The second dataset was created to study the effect of relative sizing in image classification in a set consisting of lionesses and kittens as mentioned earlier. In this case, 300 images were initially considered for each class with 240 images for training and 60 images for validation; additionally 15 images pertaining to each class were used for testing. Prior to enforcing proportional sizing as already explained in the foregoing, training and testing accuracies of 94.88% and 53.33% respectively were achieved.



(a)

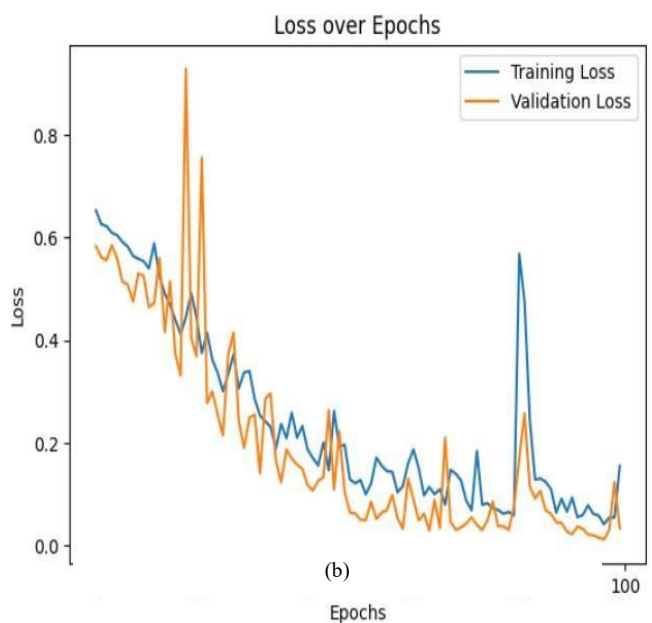
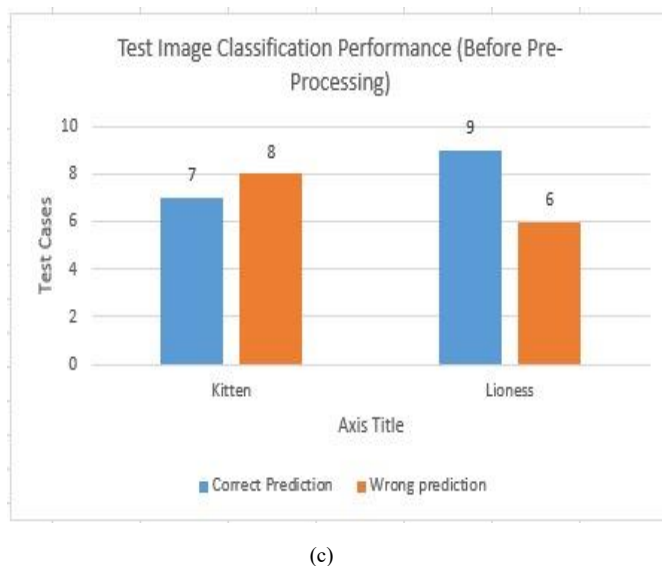
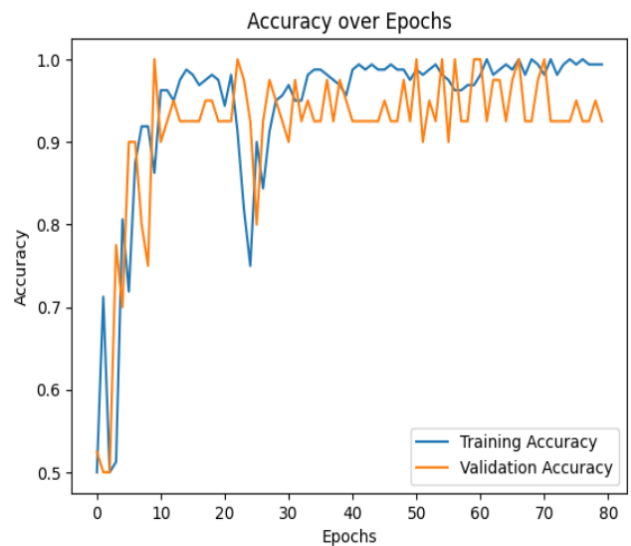
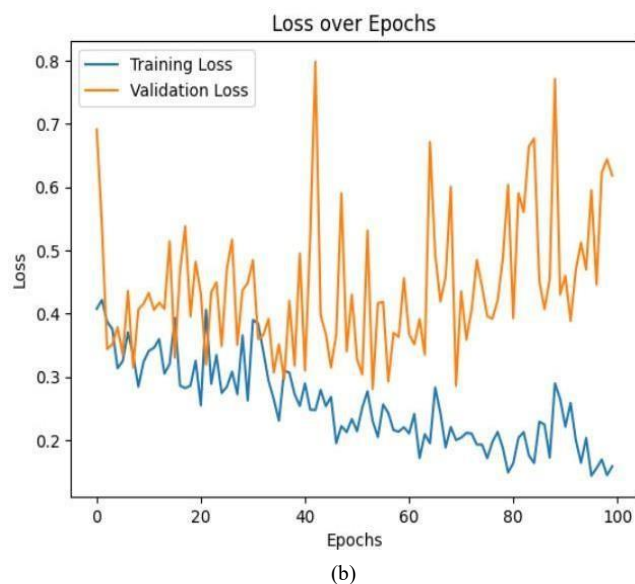


Fig. 10. (a) Accuracy vs. Epochs, and (b) Loss vs. Epochs during training for dataset (ii) before PR; (c) accuracy of testing for the trained CNN

The graphs of progression of training accuracy and loss with respect to epochs are shown in Fig. 10 (a) and (b) respectively. With all CNN parameters being the same, the proportionally resized dataset of lionesses and kittens (as shown in Fig. 4) gave rise to enhanced training and testing accuracies of 98.97% and 70% respectively.

The results mentioned are elaborated in Fig. 11; specifically, Fig. 11 (a) and (b) are respectively accuracy vs. epochs and loss vs. epochs graphs during progression of training through deep learning, and Fig. 11 (c) gives the break-up of the numbers of correctly and wrongly classified images during testing for the two classes of images considered namely, lionesses and kittens.

The results given above clearly demonstrate the increase in image classification accuracy for dataset (ii) due to proportional resizing of images of two classes which otherwise share similarities in overall geometric forms as they belong to the same family of cats.

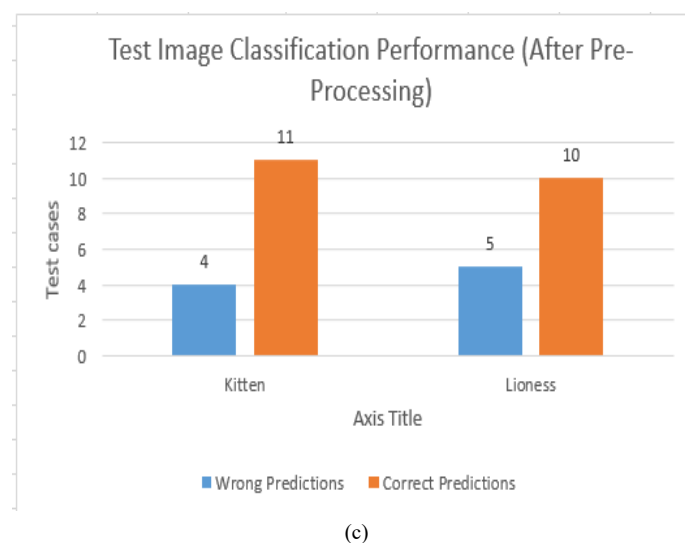


Fig. 11. (a) Accuracy vs. Epochs, and (b) Loss vs. Epochs during training for dataset (ii) after PR; (c) accuracy of testing for the trained CNN

V. CONCLUSION

The current study has dealt with the capabilities of CNN in image classification using natural and artificial (i.e. human-engineered) forms. Using a dataset containing two man-made forms i.e. cars and trucks, and a natural form viz. roses, it has been shown that higher success is achieved, during testing, by a CNN in classifying images of the former types (i.e. cars and trucks) as compared to the latter variety i.e. roses. This outcome is attributable to the fact that a natural geometric entity as typified by roses has intrinsically more variations in forms and disposition within its class whereas geometries of man-made products such as cars and trucks are more well-defined and limited. A parametric study was carried out by varying key parameters related to deep learning and it was found that in all cases, CNN consistently under-performed in detecting roses for classification. The adequacy of training dataset size used has also been established. In addition to the observations mentioned, proportional sizing, has been shown to be a key differentiator between images (such as of lions and kittens) which would have otherwise been more difficult to unambiguously pinpoint. The present study thus has provided relevant insights into the capabilities of CNN in the area of AI/ML-based image-classification, which to the authors' best knowledge, have not been reported in open literature.

VI. FUTURE WORK

Future endeavor in the current area may help in negating some of the weaknesses that have been found in CNN-based image classification through reliable and automated pre-processing of training data, and exploring efficient alternatives to deep learning.

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